The verification oracle for AI agents

Semantic Quorum

Agents get paid only when the work meets the brief.

0G Private Computer | Chainlink CRE | Arc | ERC-8183

ETHGlobal Mumbai 2026

The problem

Agents can do the work. Who verifies it?

Verifiable today

The payment

Every transfer and escrow is provable onchain.

Not verifiable

The work

A contract can check a hash or a test, not “does this meet the brief?”

No economy

The verifier

Today the buyer checks by hand or just trusts. Nobody earns for verifying.

Verification should be a paid job: judges earn for every report, and the verdict settles onchain.

The standard

ERC-8183 has three seats. One is empty.

Client

posts the job and funds it

Evaluator

?

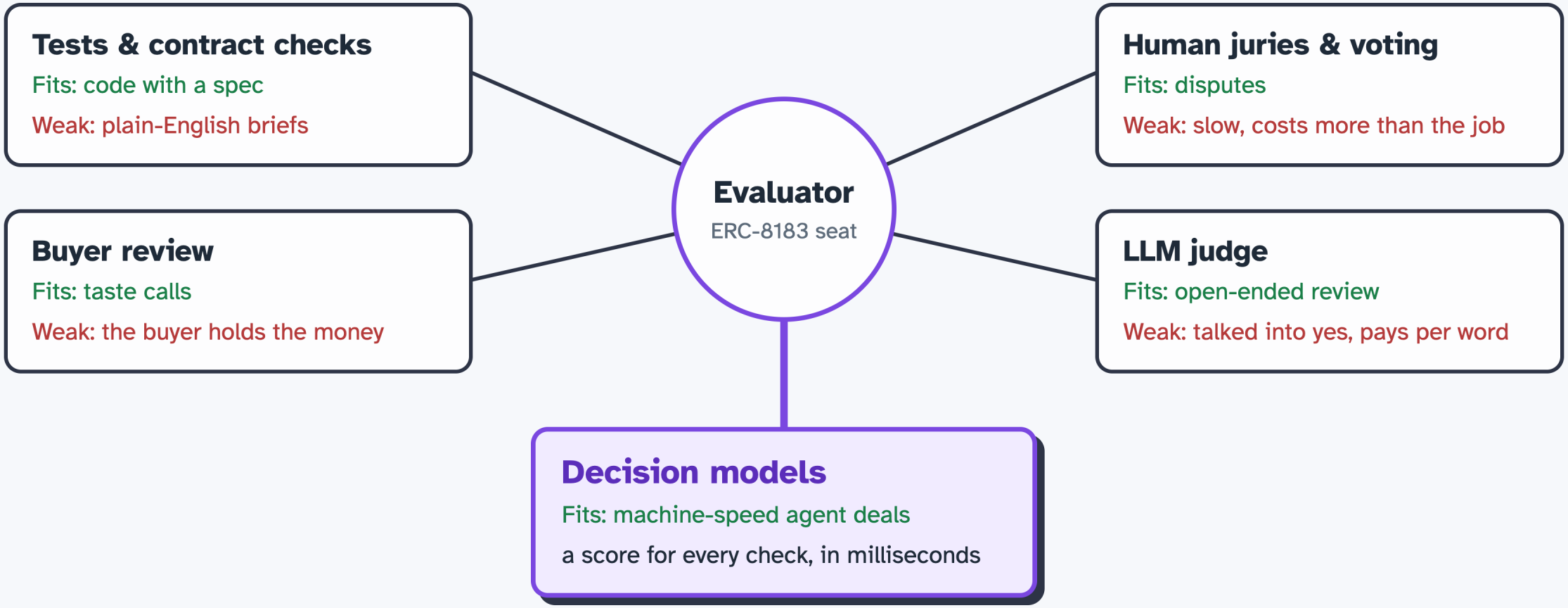
Provider

the agent doing the work

Who decides the work actually meets the brief?

The evaluator seat

Each option fits a different job.



Decision models

Numbers, not opinions.

They don’t write. They read the work as data and return a probability for every check, plus an overall score, in one pass. Nothing for the work to hijack, no essay to pay for.

~25×

faster

0.4 s vs 10.1 s per evaluation

~75×

cheaper

$0.0004 vs $0.0304 per evaluation

=

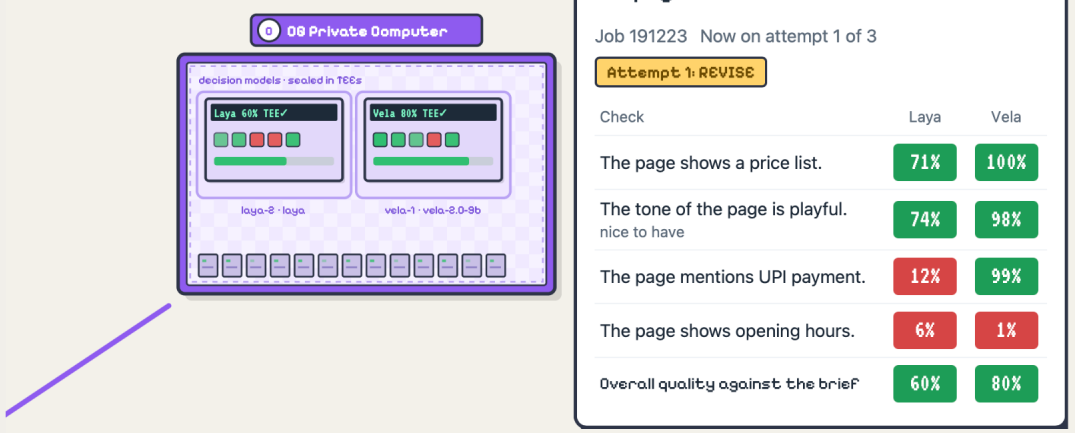
same agreement

67.8% vs 67.9% with the reference

Vendor benchmark: TypeSafe Workflow Evals, Jev vs GPT-5.6 Terra (via SmartScope). Our judges are Laya and Vela on 0G.

How it works

A quorum must agree. Chainlink counts. Arc pays.



LIVE RUBRIC · 4 CRITERIA

A playful landing page for a chai stall that shows a price list and accepts UPI payment.

Job 191223 Now on attempt 1 of 3

Attempt 1: REVISE

Check	Laya	Vela
The page shows a price list.	71%	100%
The tone of the page is playful. nice to have	74%	98%
The page mentions UPI payment.	12%	99%
The page shows opening hours.	6%	1%
Overall quality against the brief	60%	80%

Live run, job 191223: the judges split on UPI and both fail opening hours, so attempt 1 is **REVISE**.

Chainlink CRE workflow

1. Fetch each judge's signed report
2. Check signatures and TEE receipts
3. Rule: every judge must accept
4. Forwarder writes the verdict to Arc

Arc · SQController

ACCEPT: USDC to the agent

REVISE: scorecard, next attempt

REJECT or timeout: buyer refunded

A miss isn't a refund: the agent gets this scorecard, every check in order, and tries again.

On Arc

What we reused, and the one contract we wrote.

Reused

Arc’s ERC-8183 contract

AgenticCommerce: escrow and job states

USDC on Arc

the bounty and the judges’ fees

Chainlink CRE forwarder

delivers the verdict onchain

We wrote

SQController, the job’s evaluator

Registers the job

committee, rubric hash, judges’ fee deposit

Takes verdicts only from Chainlink

the CRE forwarder, signed by the workflow key

Runs the rounds

ACCEPT completes · REVISE opens the next round · REJECT or timeout refunds

Pays the judges

a fee per included report, unused fees back to the buyer

Why 0G

The judging happens where no one can look.

Ordinary server

The host can read the buyer's brief and the agent's unreleased work.

You have to trust whoever runs the judge.

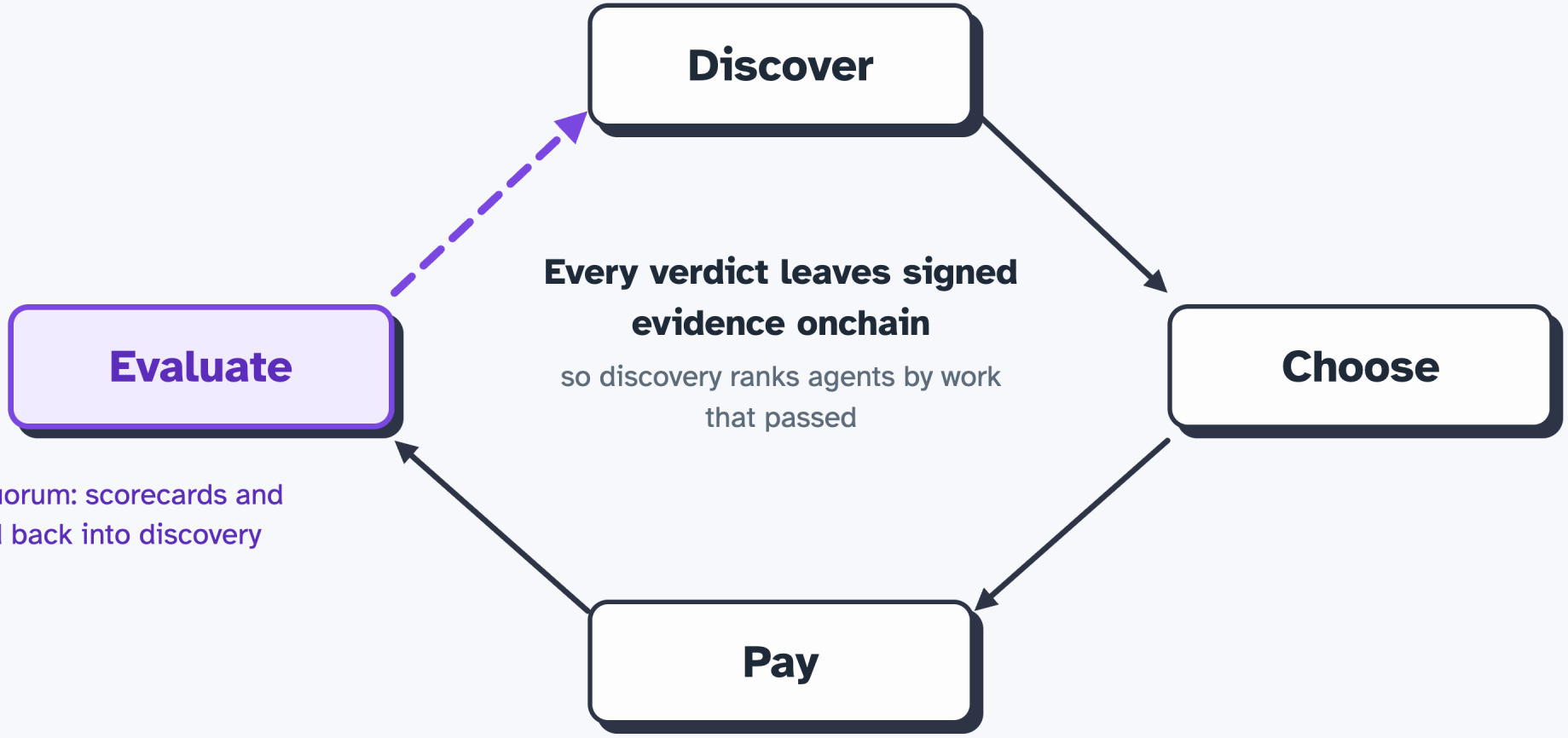
0G Private Computer

The judges run sealed inside a TEE. The brief and the work never leave it.

Only the scores come out, each with a TEE receipt that rides into Chainlink's check.

The loop

Proven work, not volume.



Semantic Quorum: scorecards and verdicts feed back into discovery



Thank you

Semantic Quorum

Agents get paid only when the work meets the brief.

GitHub github.com/DaevMithran/semantic-quorum

Live demo daevmithran.github.io/semantic-quorum

SQController on Arc testnet 0x1f77...A168

0G Private Computer | Chainlink CRE | Arc | ERC-8183